



Green University of Bangladesh

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Smart Attendance Management System with OTP-Based Self Check-In & Geo-Location Verification for GUB

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Chapter 1

Introduction

1.1 Overview

This project is a web-based **Smart Attendance System**. It is developed using PHP, JavaScript, HTML/CSS, and MySQL (MariaDB through XAMPP). The system supports student registration and login, faculty login, course creation, OTP-based attendance marking, location validation, and attendance report generation.

1.2 Motivation

Manual attendance systems are slow, error-prone, and easy to manipulate. This project was developed to address common issues in classroom attendance by combining OTP verification and location-based validation in a simple local web application:

- Reduce time spent on manual roll calls.
- Prevent proxy attendance using OTP and device checks.
- Provide instant attendance history and reports.

1.3 Problem Definition

1.3.1 Problem Statement

Traditional attendance methods do not scale well for large classes and are vulnerable to proxy presence. A reliable system needs secure verification, quick marking, and accurate reporting. This project addresses these issues by integrating OTP-based attendance, geo-location checks, and automated report generation in a lightweight web application.

1.3.2 Complex Engineering Problem (CEP) Attributes

Table 1.1 summarizes how the project maps to CEP attributes.

Table 1.1: CEP Attribute Mapping for This Project

P Attribute	How Addressed in This Project
P1: Depth of knowledge required	Requires web programming knowledge (PHP, JavaScript, SQL) and client-server interaction design.
P2: Range of conflicting requirements	Balances ease of marking with security (OTP, location, device checks).
P3: Depth of analysis required	Needs analysis of OTP time window, location rules, and attendance constraints.
P4: Familiarity of issues	Includes common web issues: input handling, SQL query design, and session logic.
P5: Extent of applicable codes	Uses PHP (backend), SQL (persistence), JavaScript (geolocation), and HTML/CSS (frontend).
P6: Extent of stakeholder involvement	Intended for students, faculty, and administrators in academic settings.
P7: Interdependence	UI, OTP generation, location data, and reporting are tightly connected.

1.4 Design Goals/Objectives

The major objectives of this project are:

- Build a role-based login system for students and faculty.
- Allow student registration and secure session handling.
- Enable faculty to create courses and generate OTPs per class.
- Validate attendance using OTP and geo-location proximity.
- Prevent duplicate attendance and multi-student device misuse.
- Provide attendance history for students and reports for faculty.

1.5 Application

This project can be used in:

- University classroom attendance management.
- Web programming lab demonstrations.
- OTP-based verification practice tasks.
- Attendance data reporting and analytics practice.

Chapter 2

Design/Development/Implementation of the Project

2.1 Introduction

This chapter explains the complete design and implementation of the Smart Attendance System project based on the actual project files.

2.2 Technologies Used

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** PHP
- **Database:** MariaDB/MySQL (XAMPP)
- **Web Server:** Apache (XAMPP)
- **Database Tool:** phpMyAdmin
- **Operating System:** Windows

2.3 Project File Structure

- `database.php` — database creation and table setup
- `connection.php` — database connection file
- `login.php` — login page and student registration
- `student.php` — student panel and attendance history
- `faculty.php` — faculty panel, course creation, OTP generation
- `mark_attendance.php` — attendance marking validation and insert

- `faculty_report.php` — attendance report generation
- `logout.php` — logout and session destroy
- `style.css` — shared UI styles

2.4 Database Design

The project uses five tables:

- `students` (`student_id`, `name`, `email`, `password`)
- `faculty` (`faculty_id`, `name`, `email`, `password`)
- `courses` (`id`, `faculty_id`, `course_name`, `course_code`, `created_at`)
- `otp_table` (`id`, `otp_code`, `created_by`, `course_id`, `create_time`, `expire_time`, `is_active`, `faculty_lat`, `faculty_long`)
- `attendance` (`id`, `student_id`, `otp_id`, `attendance_time`, `student_lat`, `student_long`, `device_identifier`)

2.4.1 SQL Setup Script

```

1 CREATE DATABASE IF NOT EXISTS attendance_management_system;
2 USE attendance_management_system;
3
4 CREATE TABLE IF NOT EXISTS students (
5     student_id VARCHAR(50) PRIMARY KEY,
6     name VARCHAR(100),
7     email VARCHAR(100) UNIQUE,
8     password VARCHAR(100)
9 );
10
11 CREATE TABLE IF NOT EXISTS faculty (
12     faculty_id INT AUTO_INCREMENT PRIMARY KEY,
13     name VARCHAR(100),
14     email VARCHAR(100) UNIQUE,
15     password VARCHAR(100)
16 );
17
18 CREATE TABLE IF NOT EXISTS courses (
19     id INT AUTO_INCREMENT PRIMARY KEY,
20     faculty_id INT,
21     course_name VARCHAR(150) UNIQUE,
22     course_code VARCHAR(50) UNIQUE,
23     created_at DATETIME
24 );
25
26 CREATE TABLE IF NOT EXISTS otp_table (
27     id INT AUTO_INCREMENT PRIMARY KEY,
28     otp_code VARCHAR(10),
29     created_by INT,

```

```

30     course_id INT,
31     create_time DATETIME,
32     expire_time DATETIME,
33     is_active TINYINT(1),
34     faculty_lat VARCHAR(50),
35     faculty_long VARCHAR(50)
36 );
37
38 CREATE TABLE IF NOT EXISTS attendance (
39     id INT AUTO_INCREMENT PRIMARY KEY,
40     student_id VARCHAR(50),
41     otp_id INT,
42     attendance_time DATETIME,
43     student_lat VARCHAR(50),
44     student_long VARCHAR(50),
45     device_identifier VARCHAR(100),
46     UNIQUE KEY unique_student_otp (student_id, otp_id)
47 );

```

Listing 2.1: Database and Table Setup Script

2.5 Workflow Summary

- Student registers and logs in; faculty logs in with seeded account.
- Faculty creates courses and generates OTP for a class.
- OTP is time-limited and linked to a course.
- Student selects course, enters OTP, and submits attendance.
- System validates OTP, time, location, and device rules.
- Attendance record is saved and shown in student history.
- Faculty generates report with per-class attendance percentage.

2.6 All Source Code

2.6.1 database.php

```

1 <?php
2 $servername = "localhost";
3 $username = "root";
4 $password = "";
5
6 $conn = mysqli_connect($servername, $username, $password);
7
8 if (!$conn) {
9     die("Connection failed");
10 }
11

```



```

12 $sql = "CREATE DATABASE IF NOT EXISTS attendance_management_system
    ";
13 if ($conn->query($sql) === TRUE) {
14     echo "Database created successfully<br>";
15 } else {
16     echo "Error creating database: " . $conn->error . "<br>";
17 }
18
19 $conn = mysqli_connect($servername, $username, $password, "
    attendance_management_system");
20
21 if (!$conn) {
22     die("Database connect failed");
23 }
24
25 $sql1 = "CREATE TABLE IF NOT EXISTS students (
26     student_id VARCHAR(50) PRIMARY KEY,
27     name VARCHAR(100),
28     email VARCHAR(100) UNIQUE,
29     password VARCHAR(100)
30 )";
31 $conn->query($sql1);
32
33 $sql2 = "CREATE TABLE IF NOT EXISTS faculty (
34     faculty_id INT AUTO_INCREMENT PRIMARY KEY,
35     name VARCHAR(100),
36     email VARCHAR(100) UNIQUE,
37     password VARCHAR(100)
38 )";
39 $conn->query($sql2);
40
41 $sql3 = "CREATE TABLE IF NOT EXISTS courses (
42     id INT AUTO_INCREMENT PRIMARY KEY,
43     faculty_id INT,
44     course_name VARCHAR(150) UNIQUE,
45     course_code VARCHAR(50) UNIQUE,
46     created_at DATETIME
47 )";
48 $conn->query($sql3);
49
50 $sql4 = "CREATE TABLE IF NOT EXISTS otp_table (
51     id INT AUTO_INCREMENT PRIMARY KEY,
52     otp_code VARCHAR(10),
53     created_by INT,
54     course_id INT,
55     create_time DATETIME,
56     expire_time DATETIME,
57     is_active TINYINT(1),
58     faculty_lat VARCHAR(50),
59     faculty_long VARCHAR(50)
60 )";
61 $conn->query($sql4);
62
63 mysqli_query($conn, "ALTER TABLE otp_table ADD COLUMN faculty_lat
    VARCHAR(50)");
64 mysqli_query($conn, "ALTER TABLE otp_table ADD COLUMN faculty_long
    VARCHAR(50)");
65

```

```

66 $sql5 = "CREATE TABLE IF NOT EXISTS attendance (
67 id INT AUTO_INCREMENT PRIMARY KEY,
68 student_id VARCHAR(50),
69 otp_id INT,
70 attendance_time DATETIME,
71 student_lat VARCHAR(50),
72 student_long VARCHAR(50),
73 device_identifier VARCHAR(100),
74 UNIQUE KEY unique_student_otp (student_id, otp_id)
75 );";
76 $conn->query($sql5);
77
78 mysqli_query($conn, "ALTER TABLE attendance ADD COLUMN student_lat
    VARCHAR(50)");
79 mysqli_query($conn, "ALTER TABLE attendance ADD COLUMN
    student_long VARCHAR(50)");
80 mysqli_query($conn, "ALTER TABLE attendance ADD COLUMN
    device_identifier VARCHAR(100)");
81
82 $check_faculty = mysqli_query($conn, "SELECT * FROM faculty WHERE
    email='faculty@gub.com'");
83 if (mysqli_num_rows($check_faculty) == 0) {
84     mysqli_query($conn, "INSERT INTO faculty(name, email, password
    ) VALUES ('Faculty One', 'faculty@gub.com', '123456')");
85 }
86
87 echo "Tables ready and default faculty inserted (if not exists).<
    br>";
88 echo "Now open login.php";
89 ?>

```

Listing 2.2: database.php

2.6.2 connection.php

```

1 <?php
2 $servername = "localhost";
3 $username = "root";
4 $password = "";
5 $dbname = "attendance_management_system";
6
7 $conn = mysqli_connect($servername, $username, $password, $dbname)
    ;
8
9 if ($conn) {
10     // connected
11 } else {
12     die("Connection failed");
13 }
14 ?>

```

Listing 2.3: connection.php

2.6.3 login.php

```

1 <?php
2 session_start();
3 include("connection.php");
4
5 $message = "";
6
7 if (isset($_POST['login_submit'])) {
8     $role = $_POST['role'];
9     $email = trim($_POST['email']);
10    $password = trim($_POST['password']);
11
12    if ($role == "student") {
13        $sql = "SELECT * FROM students WHERE email='$email' AND
password='$password'";
14        $result = mysqli_query($conn, $sql);
15        if (mysqli_num_rows($result) == 1) {
16            $row = mysqli_fetch_assoc($result);
17            $_SESSION['role'] = 'student';
18            $_SESSION['student_id'] = $row['student_id'];
19            $_SESSION['student_name'] = $row['name'];
20            header("Location: student.php");
21            exit();
22        } else {
23            $message = "Invalid student login";
24        }
25    } else if ($role == "faculty") {
26        $sql = "SELECT * FROM faculty WHERE email='$email' AND
password='$password'";
27        $result = mysqli_query($conn, $sql);
28        if (mysqli_num_rows($result) == 1) {
29            $row = mysqli_fetch_assoc($result);
30            $_SESSION['role'] = 'faculty';
31            $_SESSION['faculty_id'] = $row['faculty_id'];
32            $_SESSION['faculty_name'] = $row['name'];
33            header("Location: faculty.php");
34            exit();
35        } else {
36            $message = "Invalid faculty login";
37        }
38    }
39 }
40
41 if (isset($_POST['register_submit'])) {
42     $student_id = trim($_POST['student_id']);
43     $name = trim($_POST['name']);
44     $reg_email = trim($_POST['reg_email']);
45     $reg_password = trim($_POST['reg_password']);
46
47     if ($student_id == "" || $name == "" || $reg_email == "" ||
$reg_password == "") {
48         $message = "Please fill all student registration fields";
49     } else {
50         $insert_sql = "INSERT INTO students (student_id, name,
email, password) VALUES ('$student_id', '$name', '$reg_email',
'$reg_password')";
51         $insert_result = mysqli_query($conn, $insert_sql);
52
53         if ($insert_result) {

```

```

54         $message = "Student registration successful. Now login
";
55     } else {
56         if (strpos(mysqli_error($conn), 'student_id') !==
false) {
57             $message = "Student ID already exists";
58         } else if (strpos(mysqli_error($conn), 'email') !==
false) {
59             $message = "Email already exists";
60         } else {
61             $message = "Registration failed";
62         }
63     }
64 }
65 }
66 ?>
67 <!DOCTYPE html>
68 <html>
69 <head>
70     <title>Attendance Login</title>
71     <link rel="stylesheet" href="style.css">
72 </head>
73 <body>
74 <div class="container">
75     <h1>Attendance Login</h1>
76
77     <?php if ($message != "") { ?>
78         <p class="message"><?php echo $message; ?></p>
79     <?php } ?>
80
81     <h3>Login As</h3>
82     <form method="POST">
83         <select name="role" required>
84             <option value="student">Student</option>
85             <option value="faculty">Faculty</option>
86         </select>
87
88         <label>Email</label>
89         <input type="email" name="email" required>
90
91         <label>Password</label>
92         <input type="password" name="password" required>
93
94         <button type="submit" name="login_submit" class="blue-btn"
>Login</button>
95     </form>
96
97     <h2>New Student Registration</h2>
98     <form method="POST">
99         <label>Student ID</label>
100         <input type="text" name="student_id" required>
101
102         <label>Full Name</label>
103         <input type="text" name="name" required>
104
105         <label>Email</label>
106         <input type="email" name="reg_email" required>
107

```

```

108     <label>Password</label>
109     <input type="password" name="reg_password" required>
110
111     <button type="submit" name="register_submit" class="green-
112     btn">Register Student</button>
113 </form>
114
115     <p class="note">Default faculty (seeded): faculty@gub.com /
116     123456</p>
117 </div>
</body>
</html>

```

Listing 2.4: login.php

2.6.4 faculty.php

```

1 <?php
2 session_start();
3 include("connection.php");
4
5 if (!isset($_SESSION['role']) || $_SESSION['role'] != 'faculty') {
6     header("Location: login.php");
7     exit();
8 }
9
10 $faculty_id = $_SESSION['faculty_id'];
11 $message = "";
12 $current_otp_data = null;
13
14 if (isset($_POST['add_course_submit'])) {
15     $course_name = trim($_POST['course_name']);
16     $course_code = trim($_POST['course_code']);
17
18     if ($course_name == "" || $course_code == "") {
19         $message = "Please fill course name and course code";
20     } else {
21         $created_at = date("Y-m-d H:i:s");
22         $insert_course = "INSERT INTO courses (faculty_id,
23         course_name, course_code, created_at) VALUES ('$faculty_id', '
24         $course_name', '$course_code', '$created_at')";
25         $result_course = mysqli_query($conn, $insert_course);
26
27         if ($result_course) {
28             $message = "Course added successfully";
29         } else {
30             if (strpos(mysqli_error($conn), 'course_name') !==
31             false) {
32                 $message = "Course name already exists";
33             } else if (strpos(mysqli_error($conn), 'course_code')
34             !== false) {
35                 $message = "Course code already exists";
36             } else {
37                 $message = "Failed to add course";
38             }
39         }
40     }
41 }

```

```

36     }
37 }
38
39 if (isset($_POST['generate_otp_submit'])) {
40     $course_id = $_POST['course_id'];
41     $faculty_lat = $_POST['faculty_lat'];
42     $faculty_long = $_POST['faculty_long'];
43
44     if ($course_id == "") {
45         $message = "Please select a course";
46     } else {
47         $check_course = mysqli_query($conn, "SELECT * FROM courses
48 WHERE id='$course_id' AND faculty_id='$faculty_id'");
49         if (mysqli_num_rows($check_course) == 0) {
50             $message = "Invalid course selected";
51         } else if ($faculty_lat == "" || $faculty_long == "") {
52             $message = "Location not found. Please allow location
53 and try again";
54         } else {
55             mysqli_query($conn, "UPDATE otp_table SET is_active=0
56 WHERE course_id='$course_id' AND is_active=1");
57
58             $otp_code = rand(100000, 999999);
59             $create_time = date("Y-m-d H:i:s");
60             $expire_time = date("Y-m-d H:i:s", strtotime("+2
61 minutes"));
62
63             $insert_otp = "INSERT INTO otp_table (otp_code,
64 created_by, course_id, create_time, expire_time, is_active,
65 faculty_lat, faculty_long) VALUES ('$otp_code', '$faculty_id',
66 '$course_id', '$create_time', '$expire_time', 1, '$faculty_lat
67 ', '$faculty_long')";
68             $otp_result = mysqli_query($conn, $insert_otp);
69
70             if ($otp_result) {
71                 $message = "OTP generated successfully";
72                 $new_otp_id = mysqli_insert_id($conn);
73                 $otp_query = mysqli_query($conn, "SELECT o.*, c.
74 course_name, c.course_code FROM otp_table o JOIN courses c ON o
75 .course_id=c.id WHERE o.id='$new_otp_id'");
76                 $current_otp_data = mysqli_fetch_assoc($otp_query)
77 ;
78                 } else {
79                     $message = "Failed to generate OTP";
80                 }
81             }
82         }
83     }
84 }
85
86 $my_courses = mysqli_query($conn, "SELECT * FROM courses WHERE
87 faculty_id='$faculty_id' ORDER BY id DESC");
88
89 if ($current_otp_data == null) {
90     $latest_active_otp = mysqli_query($conn, "SELECT o.*, c.
91 course_name, c.course_code FROM otp_table o JOIN courses c ON o
92 .course_id=c.id WHERE o.created_by='$faculty_id' AND o.
93 is_active=1 ORDER BY o.id DESC LIMIT 1");
94     if (mysqli_num_rows($latest_active_otp) > 0) {

```

```

79         $current_otp_data = mysqli_fetch_assoc($latest_active_otp)
80     ;
81 }
82 ?>
83 <!DOCTYPE html>
84 <html>
85 <head>
86     <title>Faculty Panel</title>
87     <link rel="stylesheet" href="style.css">
88 </head>
89 <body>
90 <div class="container wide">
91     <h1>Faculty Panel</h1>
92     <p>Welcome <?php echo $_SESSION['faculty_name']; ?></p>
93     <p><a href="faculty_report.php">Attendance Report</a> | <a
href="logout.php">Logout</a></p>
94
95     <?php if ($message != "") { ?>
96         <p class="message"><?php echo $message; ?></p>
97     <?php } ?>
98
99     <h2>Add Course</h2>
100     <form method="POST">
101         <label>Course Name</label>
102         <input type="text" name="course_name" placeholder="Example
: Web Programming" required>
103
104         <label>Course Code</label>
105         <input type="text" name="course_code" placeholder="Example
: CSE 301" required>
106
107         <button type="submit" name="add_course_submit" class="
green-btn">Add Course</button>
108     </form>
109
110     <h2>Generate Course OTP</h2>
111     <form method="POST">
112         <label>Select Course</label>
113         <select name="course_id" required>
114             <option value="">Choose a course</option>
115             <?php
116                 $course_dropdown = mysqli_query($conn, "SELECT * FROM
courses WHERE faculty_id='<?php echo $faculty_id'; ?>' ORDER BY id DESC");
117                 while ($course = mysqli_fetch_assoc($course_dropdown))
118                 {
119                     <?>
120                     <option value="<?php echo $course['id']; ?>"><?php
echo $course['course_code']; ?> - <?php echo $course['
course_name']; ?></option>
121                     <?php } ?>
122                 </select>
123
124                 <input type="hidden" name="faculty_lat" id="faculty_lat">
125                 <input type="hidden" name="faculty_long" id="faculty_long">
126
127                 <button type="submit" name="generate_otp_submit" class="

```

```

127     green-btn">Generate OTP</button>
128 </form>
129
130     <?php if ($current_otp_data != null) { ?>
131         <h2>Current OTP</h2>
132         <p>Course: <?php echo $current_otp_data['course_code']; ?>
133         - <?php echo $current_otp_data['course_name']; ?></p>
134         <p>OTP Code: <strong><?php echo $current_otp_data['
135         otp_code']; ?></strong></p>
136         <p>Create Time: <?php echo $current_otp_data['create_time'
137         ]; ?></p>
138         <p>Expire Time: <?php echo $current_otp_data['expire_time'
139         ]; ?></p>
140     <?php } ?>
141
142     <h2>My Courses</h2>
143     <table>
144         <tr>
145             <th>Course Code</th>
146             <th>Course Name</th>
147             <th>Created At</th>
148         </tr>
149         <?php while ($row = mysqli_fetch_assoc($my_courses)) { ?>
150             <tr>
151                 <td><?php echo $row['course_code']; ?></td>
152                 <td><?php echo $row['course_name']; ?></td>
153                 <td><?php echo $row['created_at']; ?></td>
154             </tr>
155         <?php } ?>
156     </table>
157 </div>
158
159 <script>
160 if (navigator.geolocation) {
161     navigator.geolocation.getCurrentPosition(function(position) {
162         document.getElementById("faculty_lat").value = position.
163         coords.latitude;
164         document.getElementById("faculty_long").value = position.
165         coords.longitude;
166     });
167 }
168 </script>
169 </body>
170 </html>

```

Listing 2.5: faculty.php

2.6.5 student.php

```

1 <?php
2 session_start();
3 include("connection.php");
4
5 if (!isset($_SESSION['role']) || $_SESSION['role'] != 'student') {
6     header("Location: login.php");
7     exit();

```



```

8 }
9
10 $message = "";
11 if (isset($_GET['msg'])) {
12     $message = $_GET['msg'];
13 }
14
15 $courses = mysqli_query($conn, "SELECT * FROM courses ORDER BY id
    DESC");
16 $student_id = $_SESSION['student_id'];
17
18 $history = mysqli_query($conn, "SELECT a.attendance_time, c.
    course_code, c.course_name, o.otp_code
19 FROM attendance a
20 JOIN otp_table o ON a.otp_id=o.id
21 JOIN courses c ON o.course_id=c.id
22 WHERE a.student_id='$student_id'
23 ORDER BY a.id DESC");
24 ?>
25 <!DOCTYPE html>
26 <html>
27 <head>
28     <title>Student Panel</title>
29     <link rel="stylesheet" href="style.css">
30 </head>
31 <body>
32 <div class="container wide">
33     <h1>Student Panel</h1>
34     <p>Welcome <?php echo $_SESSION['student_name']; ?> (<?php
    echo $_SESSION['student_id']; ?>)</p>
35     <p><a href="logout.php">Logout</a></p>
36
37     <?php if ($message != "") { ?>
38         <p class="message"><?php echo $message; ?></p>
39     <?php } ?>
40
41     <h2>Mark Attendance</h2>
42     <form method="POST" action="mark_attendance.php">
43         <label>Select Course</label>
44         <select name="course_id" required>
45             <option value="">Choose a course</option>
46             <?php while ($course = mysqli_fetch_assoc($courses)) {
    ?>
47                 <option value="<?php echo $course['id']; ?>"><?php
    echo $course['course_code']; ?> - <?php echo $course['
    course_name']; ?></option>
48             <?php } ?>
49         </select>
50
51         <label>Enter OTP</label>
52         <input type="text" name="otp_code" maxlength="6" required>
53
54         <input type="hidden" name="student_lat" id="student_lat">
55         <input type="hidden" name="student_long" id="student_long"
    >
56
57         <button type="submit" name="mark_submit" class="blue-btn">
    Submit Attendance</button>

```

```

58     </form>
59
60     <h2>My Attendance History</h2>
61     <table>
62         <tr>
63             <th>Course</th>
64             <th>OTP</th>
65             <th>Attendance Time</th>
66         </tr>
67         <?php while ($row = mysqli_fetch_assoc($history)) { ?>
68             <tr>
69                 <td><?php echo $row['course_code']; ?> - <?php echo
$row['course_name']; ?></td>
70                 <td><?php echo $row['otp_code']; ?></td>
71                 <td><?php echo $row['attendance_time']; ?></td>
72             </tr>
73         <?php } ?>
74     </table>
75 </div>
76
77 <script>
78 if (navigator.geolocation) {
79     navigator.geolocation.getCurrentPosition(function(position) {
80         document.getElementById("student_lat").value = position.
coords.latitude;
81         document.getElementById("student_long").value = position.
coords.longitude;
82     });
83 }
84 </script>
85 </body>
86 </html>

```

Listing 2.6: student.php

2.6.6 mark_attendance.php

```

1 <?php
2 session_start();
3 include("connection.php");
4
5 if (!isset($_SESSION['role']) || $_SESSION['role'] != 'student') {
6     header("Location: login.php");
7     exit();
8 }
9
10 if (!isset($_POST['mark_submit'])) {
11     header("Location: student.php?msg=Invalid+request");
12     exit();
13 }
14
15 $student_id = $_SESSION['student_id'];
16 $course_id = trim($_POST['course_id']);
17 $otp_code = trim($_POST['otp_code']);
18 $student_lat = trim($_POST['student_lat']);
19 $student_long = trim($_POST['student_long']);

```

```

20 $device_identifier = $_SERVER['REMOTE_ADDR'];
21
22 if ($course_id == "" || $otp_code == "") {
23     header("Location: student.php?msg=Please+select+course+and+
24     enter+OTP");
25     exit();
26 }
27 $otp_query = mysqli_query($conn, "SELECT * FROM otp_table WHERE
28     otp_code='$otp_code' AND course_id='$course_id' AND is_active=1
29     ORDER BY id DESC LIMIT 1");
30
31 if (mysqli_num_rows($otp_query) == 0) {
32     header("Location: student.php?msg=Invalid+OTP+or+inactive+OTP");
33     exit();
34 }
35 $otp_row = mysqli_fetch_assoc($otp_query);
36 $otp_id = $otp_row['id'];
37 $faculty_lat = $otp_row['faculty_lat'];
38 $faculty_long = $otp_row['faculty_long'];
39 $expire_time = strtotime($otp_row['expire_time']);
40 $current_time = time();
41
42 if ($current_time > $expire_time) {
43     mysqli_query($conn, "UPDATE otp_table SET is_active=0 WHERE id
44     ='$otp_id'");
45     header("Location: student.php?msg=OTP+expired");
46     exit();
47 }
48
49 if ($student_lat == "" || $student_long == "" || $faculty_lat == ""
50     || $faculty_long == "") {
51     header("Location: student.php?msg=Location+not+found");
52     exit();
53 }
54
55 $lat_diff = abs($student_lat - $faculty_lat);
56 $long_diff = abs($student_long - $faculty_long);
57
58 if ($lat_diff > 0.001 || $long_diff > 0.001) {
59     header("Location: student.php?msg=You+are+outside+the+
60     classroom+boundary");
61     exit();
62 }
63
64 $check_duplicate = mysqli_query($conn, "SELECT * FROM attendance
65     WHERE student_id='$student_id' AND otp_id='$otp_id'");
66
67 if (mysqli_num_rows($check_duplicate) > 0) {
68     header("Location: student.php?msg=You+already+marked+
69     attendance+for+this+OTP");
70     exit();
71 }
72
73 $check_device = mysqli_query($conn, "SELECT * FROM attendance
74     WHERE otp_id='$otp_id' AND device_identifier='
75     $device_identifier' AND student_id!='$student_id'");
76
77 if (mysqli_num_rows($check_device) > 0) {

```

```

67     header("Location: student.php?msg=This+device+already+marked+
68     attendance+for+another+student");
69     exit();
70 }
71 $attendance_time = date("Y-m-d H:i:s");
72 $insert_attendance = mysqli_query($conn, "INSERT INTO attendance (
73     student_id, otp_id, attendance_time, student_lat, student_long,
74     device_identifier) VALUES ('$student_id', '$otp_id', '
75     $attendance_time', '$student_lat', '$student_long', '
76     $device_identifier')");
77
78 if ($insert_attendance) {
79     header("Location: student.php?msg=Attendance+marked+
80     successfully");
81 } else {
82     header("Location: student.php?msg=Failed+to+mark+attendance");
83 }
84 exit();
85 ?>

```

Listing 2.7: mark_attendance.php

2.6.7 faculty_report.php

```

1  <?php
2  session_start();
3  include("connection.php");
4
5  if (!isset($_SESSION['role']) || $_SESSION['role'] != 'faculty') {
6      header("Location: login.php");
7      exit();
8  }
9
10 $faculty_id = $_SESSION['faculty_id'];
11 $course_id = "";
12
13 if (isset($_POST['report_submit'])) {
14     $course_id = $_POST['course_id'];
15 }
16
17 $courses = mysqli_query($conn, "SELECT * FROM courses WHERE
18     faculty_id='$faculty_id' ORDER BY id DESC");
19 ?>
20 <!DOCTYPE html>
21 <html>
22 <head>
23     <title>Attendance Report</title>
24     <link rel="stylesheet" href="style.css">
25 </head>
26 <body>
27 <div class="container wide">
28     <h1>Attendance Report</h1>
29     <p><a href="faculty.php">Back to Faculty Panel</a> | <a href="
30     logout.php">Logout</a></p>

```

```

30 <form method="POST">
31     <label>Select Course</label>
32     <select name="course_id" required>
33         <option value="">Choose a course</option>
34         <?php while ($row = mysqli_fetch_assoc($courses)) { ?>
35             <option value="<?php echo $row['id']; ?>" <?php if
($course_id == $row['id']) { echo "selected"; } ?>><?php echo
$row['course_code']; ?> - <?php echo $row['course_name']; ?></
option>
36         <?php } ?>
37     </select>
38     <button type="submit" name="report_submit" class="green-
btn">Show Report</button>
39 </form>
40
41 <?php if ($course_id != "") { ?>
42     <?php
43     $otp_result = mysqli_query($conn, "SELECT * FROM otp_table
WHERE course_id='$course_id' ORDER BY create_time ASC");
44     $otp_ids = array();
45     $otp_dates = array();
46     while ($o = mysqli_fetch_assoc($otp_result)) {
47         $otp_ids[] = $o['id'];
48         $otp_dates[] = $o['create_time'];
49     }
50
51     $students_result = mysqli_query($conn, "SELECT DISTINCT s.
student_id, s.name FROM attendance a JOIN otp_table o ON a.
otp_id=o.id JOIN students s ON a.student_id=s.student_id WHERE
o.course_id='$course_id' ORDER BY s.student_id");
52     ?>
53
54     <button onclick="window.print()" class="blue-btn">Print</
button>
55
56     <table>
57         <tr>
58             <th>Student ID</th>
59             <th>Name</th>
60             <?php for ($i = 0; $i < count($otp_dates); $i++) {
?>
61                 <th><?php echo date("Y-m-d", strtotime(
$otp_dates[$i])); ?></th>
62                 <?php } ?>
63                 <th>Total Percentage</th>
64             </tr>
65
66             <?php while ($s = mysqli_fetch_assoc($students_result)
) { ?>
67                 <?php
68                 $present_count = 0;
69                 $total_classes = count($otp_ids);
70                 ?>
71                 <tr>
72                     <td><?php echo $s['student_id']; ?></td>
73                     <td><?php echo $s['name']; ?></td>
74                     <?php for ($j = 0; $j < count($otp_ids); $j++)
{ ?>

```

```

75         <?php
76         $oid = $otp_ids[$j];
77         $check = mysqli_query($conn, "SELECT *
FROM attendance WHERE student_id='" . $s['student_id'] . "' AND
otp_id='$oid'");
78         if (mysqli_num_rows($check) > 0) {
79             echo "<td>P</td>";
80             $present_count++;
81         } else {
82             echo "<td>A</td>";
83         }
84         ?>
85     <?php } ?>
86     <?php
87     if ($total_classes > 0) {
88         $percent = ($present_count /
$total_classes) * 100;
89     } else {
90         $percent = 0;
91     }
92     ?>
93     <td><?php echo round($percent, 2); ?>%</td>
94 </tr>
95 <?php } ?>
96 </table>
97 <?php } ?>
98 </div>
99 </body>
100 </html>

```

Listing 2.8: faculty_report.php

2.6.8 logout.php

```

1 <?php
2 session_start();
3 session_unset();
4 session_destroy();
5 header("Location: login.php");
6 exit();
7 ?>

```

Listing 2.9: logout.php

2.6.9 style.css

```

1 body {
2     margin: 0;
3     padding: 20px;
4     background: #dcdcdc;
5     font-family: Arial, sans-serif;
6 }
7
8 .container {
9     width: 500px;

```

```

10     margin: 0 auto;
11     background: #eeeeee;
12     padding: 25px;
13     border: 1px solid #cfcfcf;
14 }
15
16 .container.wide {
17     width: 700px;
18 }
19
20 h1, h2, h3 {
21     margin-top: 0;
22 }
23
24 label {
25     display: block;
26     margin-top: 10px;
27     margin-bottom: 5px;
28     font-size: 18px;
29 }
30
31 input, select {
32     width: 100%;
33     box-sizing: border-box;
34     padding: 10px;
35     margin-bottom: 5px;
36     border: 1px solid #9f9f9f;
37     font-size: 18px;
38 }
39
40 button {
41     width: 100%;
42     border: none;
43     color: #ffffff;
44     font-size: 32px;
45     padding: 10px;
46     margin-top: 10px;
47     cursor: pointer;
48 }
49
50 .blue-btn {
51     background: #2f80db;
52 }
53
54 .green-btn {
55     background: #008f00;
56 }
57
58 .message {
59     background: #fffde0;
60     border: 1px solid #e2d999;
61     padding: 10px;
62 }
63
64 .note {
65     margin-top: 20px;
66     color: #555555;
67 }

```

```
68
69 table {
70     width: 100%;
71     border-collapse: collapse;
72     margin-top: 10px;
73 }
74
75 th, td {
76     border: 1px solid #a8a8a8;
77     padding: 8px;
78     text-align: left;
79 }
80
81 a {
82     color: #0a4ca8;
83 }
```

Listing 2.10: style.css

Chapter 3

Performance Evaluation

3.1 Simulation Environment

The project was tested in a local development environment:

- XAMPP (Apache + MariaDB)
- PHP runtime from XAMPP
- phpMyAdmin for database inspection
- Windows operating system

3.2 Testing Strategy

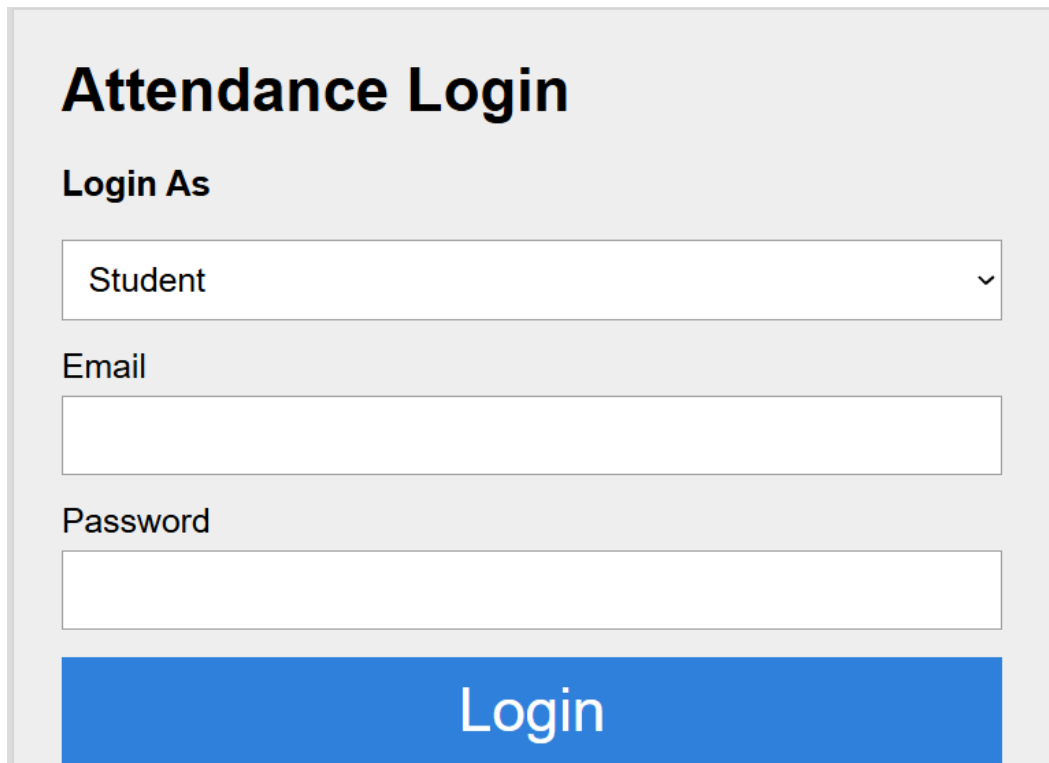
Testing was done using functional test scenarios:

- Student registration and login.
- Faculty login and course creation.
- OTP generation and validity window.
- Attendance marking with OTP and location check.
- Duplicate attendance prevention.
- Attendance history and report generation.

3.3 Results Analysis

3.3.1 Test 1: Student Registration and Login

Expected: Student account created and login redirects to student panel.



The image shows a web form titled "Attendance Login". It includes a "Login As" dropdown menu with "Student" selected, an "Email" text input field, a "Password" text input field, and a blue "Login" button.

Attendance Login

Login As

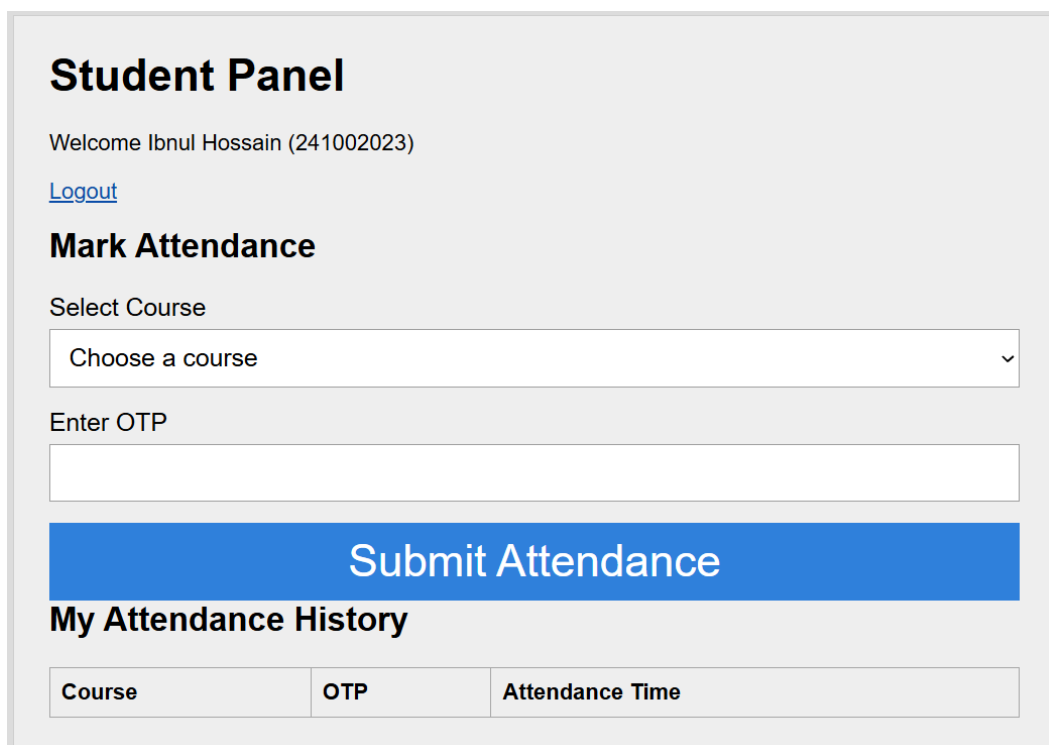
Student

Email

Password

Login

Figure 3.1: Attendance Login Page



The image shows a web panel titled "Student Panel". It displays a welcome message for "Ibnul Hossain (241002023)" with a "Logout" link. Below this is a "Mark Attendance" section with a "Select Course" dropdown menu (showing "Choose a course") and an "Enter OTP" text input field. A blue "Submit Attendance" button is present. At the bottom, there is a section titled "My Attendance History" with a table header.

Student Panel

Welcome Ibnul Hossain (241002023)

[Logout](#)

Mark Attendance

Select Course

Choose a course

Enter OTP

Submit Attendance

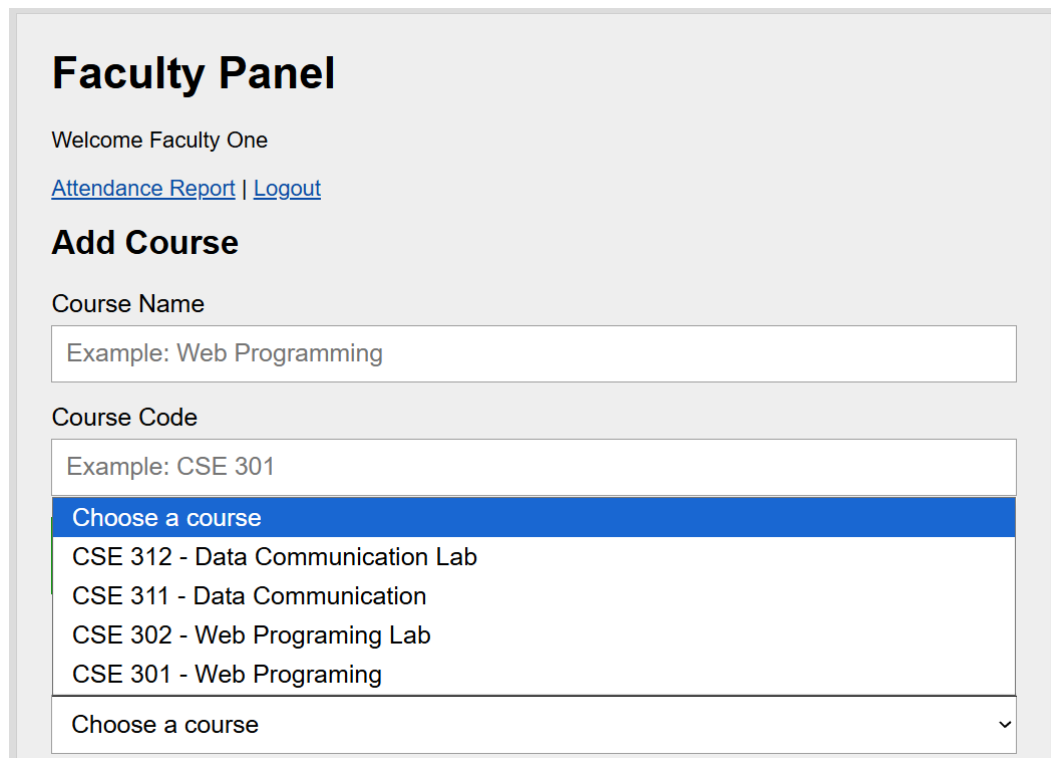
My Attendance History

Course	OTP	Attendance Time
--------	-----	-----------------

Figure 3.2: Student Panel After Login

3.3.2 Test 2: Faculty Login and Course Creation

Expected: Faculty can add a course and see it listed in the panel.

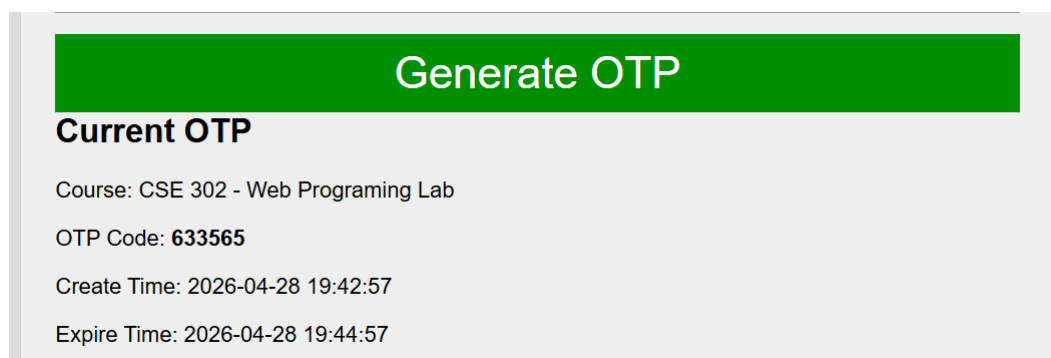


The screenshot shows a web interface titled "Faculty Panel". Below the title, it says "Welcome Faculty One" and provides links for "Attendance Report" and "Logout". The main section is titled "Add Course". It contains two input fields: "Course Name" with the placeholder text "Example: Web Programming" and "Course Code" with the placeholder text "Example: CSE 301". Below the "Course Code" field is a dropdown menu. The dropdown is currently open, showing a list of course options: "Choose a course" (highlighted in blue), "CSE 312 - Data Communication Lab", "CSE 311 - Data Communication", "CSE 302 - Web Programing Lab", and "CSE 301 - Web Programing". At the bottom of the dropdown is a "Choose a course" option with a downward arrow.

Figure 3.3: Faculty Panel with Course List

3.3.3 Test 3: OTP Generation

Expected: OTP is generated with course details and time window.



The screenshot shows a web interface titled "Generate OTP". Below the title, it says "Current OTP". It contains the following information: "Course: CSE 302 - Web Programing Lab", "OTP Code: 633565", "Create Time: 2026-04-28 19:42:57", and "Expire Time: 2026-04-28 19:44:57".

Figure 3.4: Generated OTP in Faculty Panel

The screenshot displays a web interface for a student's attendance management. At the top, the header 'Student Panel' is followed by a welcome message for 'Ibnul Hossain (241002023)' and a 'Logout' link. The main section is titled 'Mark Attendance' and contains a 'Select Course' dropdown menu with the placeholder text 'Choose a course'. Below this is an 'Enter OTP' text input field. A prominent blue button labeled 'Submit Attendance' is positioned below the input fields. At the bottom of the panel, the text 'My Attendance History' is visible.

Figure 3.5: Student Attendance Form (Course and OTP)

3.3.4 Test 4: Attendance Marking

Expected: Attendance is marked and appears in student history.

Student Panel

Welcome Ibnul Hossain (241002023)

[Logout](#)

Attendance marked successfully

Mark Attendance

Select Course

Choose a course

Enter OTP

Submit Attendance

My Attendance History

Course	OTP	Attendance Time
CSE 302 - Web Programing Lab	633565	2026-04-28 19:43:43

Figure 3.6: Attendance Marked Successfully

3.3.5 Test 5: Attendance Report

Expected: Report table shows presence/absence and percentage.

Attendance Report

[Back to Faculty Panel](#) | [Logout](#)

Select Course

CSE 302 - Web Programing Lab

Show Report

Print

Student ID	Name	2026-04-28	2026-04-28	Total Percentage
241002010	Ismum Hossain	P	A	50%
241002023	Ibnul Hossain	A	P	50%

Figure 3.7: Faculty Attendance Report Table

3.4 Result Discussion

The project successfully demonstrates end-to-end integration of OTP-based attendance, geo-location checks, and automated reporting. All major functional requirements were implemented and validated in local testing.

	id	student_id	otp_id	attendance_time	student_lat	student_long	device_identifier
☐ Edit ☐ Copy ☐ Delete	1	241002046	1	2026-04-23 23:12:26	NULL	NULL	NULL
☐ Edit ☐ Copy ☐ Delete	2	241002046	3	2026-04-28 13:56:04	23.690075189369175	90.46735411527331	::1
☐ Edit ☐ Copy ☐ Delete	3	241002010	4	2026-04-28 14:10:47	23.690083339035997	90.46737079238619	::1
☐ Edit ☐ Copy ☐ Delete	4	241002023	6	2026-04-28 19:43:43	23.69007954594978	90.46735487556778	::1

Figure 3.8: Database Tables in phpMyAdmin

Chapter 4

Conclusion

4.1 Discussion

The Smart Attendance System achieved its core goals. The application supports role-based access, OTP-based attendance, location validation, and attendance reporting. It is suitable as a compact academic project for demonstrating practical PHP and MySQL concepts.

4.2 Limitations

Current limitations are:

- Passwords are stored in plain text in the database.
- SQL queries are directly concatenated (risk of SQL injection).
- Device identifier uses IP address, which is not unique.
- No HTTPS or encryption for data in transit.

4.3 Scope of Future Work

Future improvements can include:

- Use prepared statements for database security.
- Hash user passwords and add password reset flow.
- Add OTP expiration countdown in UI.
- Improve device identification (browser fingerprint or device ID).
- Add export of attendance report to CSV/PDF.

References

- [1] PHP Documentation Team. *PHP Manual: Sessions*. The PHP Group. Available at: <https://www.php.net/manual/en/book.session.php>
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- [4] OWASP Foundation. *SQL Injection Prevention Cheat Sheet*. OWASP Cheat Sheet Series. Available at: https://cheatsheetseries.owasp.org/cheatsheets/SQL_injection_prevention_cheat_sheet.html
- [5] Apache Friends. *XAMPP Documentation*. Available at: <https://www.apachefriends.org/index.html>